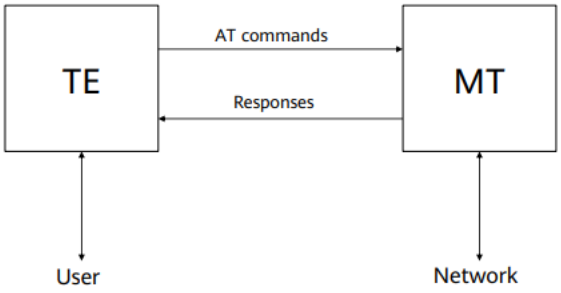
RANGKUMAN BAB 9 IOT

A. AT Command Types

AT Commands are used to control the rules of interaction TE and MT



AT Command Types

Category	Syntax	Example	Fuction
Test Command	AT+<x>=?	AT+CMEE=?	to display valid parameter values set by other AT commands
Read Command	AT+<x>?	AT+CMEE?	to query attributes set by other AT commands.
Set Command	AT+<x>=<…>	AT+CMEE=0	to instruct a module to execute a specific function
Execution Command	AT+<x>	AT+NRB	to set the attributes carried in the AT command.

B. NB-IoT AT Commands

Purpose	AT Command
Disabling a function	AT+CFUN=0
Checking the software version	AT+CGMR
Querying the international mobile equipment identity (IMEI)	AT+CGSN=1
Setting the platform address	AT+NCDP=xx.xx.xx.xx,xx
Configuring an access point name (APN)	AT+CGDCONT=1,"IP","xxxx"
Rebooting a module	AT+NRB
Enabling a function	AT+CFUN=1
Querying the international mobile subscriber identity (IMSI) of a SIM card	AT+CIMI
Notifying the terminal of connecting to the base station	AT+CSCON=1

Purpose	AT Command
Notifying the terminal of connecting to the core network	AT+CEREG=2
Notifying the terminal of downlink data transmission	AT+NNMI=1
Notifying the terminal of successful data transmission	AT+NSMI=1
Attaching to a network	AT+CGATT=1
Querying the terminal status	AT+NUESTATS
Querying the IP address assigned by the core network	AT+CGPADDR
Transmitting data	AT+NMGS=1,11
Querying the sending buffer	AT+NQMGS
Querying the receiving buffer	AT+NQMGR

Connecting an NB-IoT Device to the Huawei Cloud IoT Platform

1) Power on the terminal, and run the **AT+NRB** command to reset the terminal. If **OK** is returned, the terminal is running properly.
2) Run the **AT+NTSETID=1,DEVICEID** command to specify the device ID. The device ID is the terminal IMEI. If the command is executed, OK is returned.
3) Run the **AT+NCDP=IP,PORT** command to set the IP address and port for connecting to the IoT platform. The port is 5683. If the command is executed, OK is returned.
4) Run the **AT+CFUN=1** command to enable the network access function. If the command is executed, OK is returned.
5) Run the **AT+NBAND=Frequency_band** command to specify the frequency band. If the command is executed, OK is returned.
6) Run the **AT+CGDCONT=1,"IP","APN"** command to set the IoT core APN. If the command is executed, OK is returned.
7) Run the **AT+CGATT=1** command to connect the terminal to the network. If the command is executed, OK is returned.
8) Run the **AT+CGPADDR** command to check whether the terminal has obtained the IP address assigned by the IoT core network. If it has, the terminal has accessed the network.
9) Run the **AT+NMGS=DATASIZE,DATA** command to enable the terminal to send upstream data. If the upstream data is sent, OK is returned.
10) If the IoT platform sends downstream data to the terminal, obtain downstream data by running the AT+NMGR command.

C. Wi-Fi AT Commands

Universal Wi-Fi AT Commands

Command	Description
AT	Testing the AT function
AT+HELP	Viewing available AT commands
AT+MAC	Managing the MAC address
AT+IPERF	Testing performance
AT+SYSINFO	Viewing system information
AT+PING	Testing the IPv4 network connection
AT+PING6	Testing the IPv6 network connection
AT+DNS	Setting the DNS server address of the board
AT+ARP	Setting the ARP offload
AT+SLP	Setting the system low power function
AT+PS	Setting the Wi-Fi low power function

Instruction	Description
AT+DHCP	Running the DHCP client command
AT+DHCPS	Running the DHCP server command
AT+IFCFG	Configuring an interface
AT+CC	Setting a country/region code
AT+DUMP	Reading the latest exception information
AT+NETSTAT	Checking the network status
AT+CSV	Querying the software version
AT+RST	Resetting a board
AT+WKGPIO	Setting the GPIO wakeup source in light sleep or deep sleep mode
AT+USLP	Entering the ultra-deep sleep mode
AT+SETUART	Configuring the serial port function

AP-related Wi-Fi AT Commands

As a non-3GPP short-range wireless communications technology, Wi-Fi involves different AT commands from NB-IoT. Wi-Fi AT commands are mainly used to interact with the gateway and access the network through the gateway. Carrier data related to SIM cards, wireless networks, and core networks is not involved.

Access Point Example

Purpose	AT Command
Restarting the module	AT+RST
Querying the version	AT+GMR
Scanning nearby APs	AT+CWLAP
Connecting to an AP	AT+CWJAP
Disconnecting from an AP	AT+CWQAP
Querying connection information	AT+CIPSTATUS
Resolving the domain name	AT+CIPDOMAIN

Purpose	AT Command
Establishing a connection	AT+CIPSTART
Starting transparent transmission	AT+CIPMODE
Transmitting data	AT+CIPSEND
Querying the local IP address	AT+CIFSR
Using the ping operation	AT+PING
Restoring to factory settings	AT+RESTORE
Querying the available memory space of the system	AT+SYSRAM

STA-related Wi-Fi AT Commands

A station (STA) functions as a client in a wireless local area network (WLAN). It can be fixed or mobile. It can be a computer with a wireless network adapter or a smartphone with a Wi-Fi module

Purpose	AT Command
Starting the STA	AT+STARTSTA
Stopping the STA	AT+STOPSTA
Configuring the reconnection policy	AT+RECONN
Initiating STA scanning	AT+SCAN
Scanning a specified channel	AT+SCANCHN
Scanning a specified SSID	AT+SCANSSID
Viewing scan results	AT+SCANRESULT

Purpose	AT Command
Initiating a connection to the AP	AT+CONN
Initiating a fast connection to the AP	AT+FCONN
Disconnecting from the AP	AT+DISCONN
viewing the STA status	AT+STASTAT
Using Wi-Fi protected setup (WPS) push-button configuration (PBC) for connection	AT+PBC
Using a WPS pin for connection	AT+PIN
Viewing the generated pin	AT+PINSHOW

Connecting a Wi-Fi Device to the Huawei Cloud IoT Platform

1) Power on the terminal, and run the **AT+RST** command to reset the terminal. If OK is returned, the Wi-Fi mode has been configured on the terminal.
2) Run the **AT+CWJAP="SSID","Password"** command to connect to the router. If the command is executed, OK is returned.
3) Run the **AT+CIFSR** command to query the IP address of the local device. If the command is executed, OK is returned.
4) Run the **AT+CIPSTART="TCP","IP","PORT"** command to set the IP address for connecting to the IoT platform. The port is 1883 (MQTT port). If the command is executed, OK is returned.

5) Run the **AT+CIPSEND= *<lenght>*** command to send data. After > is returned, input the data. If the command is executed, SEND OK is returned

D. AT Command for Huawei-Certified Modules

For modules with Huawei compatibility certification, the AT commands and format specifications are similar to general specifications. Some manufacturers' modules may be implemented slightly differently, due to their AT channel limitations. These differences will be stated in the special description by module manufactures.

Purpose	AT Command
Obtaining the Huawei SDK version	AT+HMVER
Setting MQTT connection parameters	AT+HMCON
Disconnecting from the Huawei Cloud IoT platform	AT+HMDIS
Sending MQTT data to a topic	AT+HMPUB
Transmitting data received by the module to an external MCU	+HMREC
Transmitting the module connection or disconnection status to an external MCU	+HMSTS
Subscribing to a custom topic	AT+HMSUB
Unsubscribing from a custom topic	AT+HMUNS
Setting a server or client certificate	AT+HMPKS

Connecting a Huawei-certified Module to the Huawei Cloud IoT Platform

1) Run **AT+HMCON=bsmode,lifetime,"serverip","serverport","deviceID","passwd",codec** to connect to the Huawei Cloud IoT platform. If **+HMCON OK** is returned, the connection is successful.
2) Run **AT+HMSUB=qos,topic** to subscribe to a custom topic. If +HMSUB OK is returned, the subscription is successful.
3) Run **AT+HMPUB=qos,topic,payload_len,payload** to report messages or properties. If +HMPUB OK is returned, the reporting is successful.
4) Deliver a command on the Huawei Cloud IoT platform. If the device receives **+HMREC:topic,payload_len,payload,** the command is received.
5) Run **AT+HMUNS="topic"** to unsubscribe from a custom topic. If +HMUNS OK is returned, the unsubscription is successful.
6) Run **AT+HMDIS** to disconnect the device from the IoT platform.
7) Run **AT+HMPKS=type,para1,[para2],"Certificate"** to set the server or client certificate

QUIZ

1) Which AT command is used to initiate a connection to the AP?

   A. AT+STOPSTA

   B. AT+SCAN

   C. AT+CONN [answer]

   D. AT+STASTAT

2) What is the Port Number of HUAWEI CLOUD MQTT?

   A. 5683

   B. 1883 [answer]

   C. 1884

   D. 443

3) Which of the following command types is "AT+CMEE?" classified into?

   A. Test command

   B. Read command [answer]

   C. Set command

   D. Execution command

4) Which of the following are types of AT commands

   A. Test command [answer]

   B. Read command [answer]

   C. Set command [answer]

   D. Execution command [answer]

5) AT commands are used to control the rules of interactions between a TE and an MT.

   ○ **Right [answer]**

   ○ Wrong